

Research Study: CS 421

Reproducing “Beemo”: A Study on Expert-Edited Machine-Generated Text Detection

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Reproducibility Summary

Scope of Reproducibility — The original paper claims that expert-edited machine-generated texts can reliably evade state-of-the-art machine-generated text (MGT) detectors, while LLM-edited texts are still often recognized as machine-generated. The authors also claim that zero-shot detectors generalize better to edited MGTs than pretrained detectors. Our objective is to reproduce these findings by evaluating the same set of zero-shot and pretrained detectors on the Beemo dataset and validating whether expert edits indeed reduce detectability.

Methodology — We used the authors’ released Beemo dataset and detector implementations (Binoculars, DetectGPT, MAGE, RADAR, etc.) to reproduce the paper’s evaluation setup. All experiments were run on Google Colab using a single GPU environment (T4), with minimal GPU hours required since detector predictions were precomputed. Our evaluation followed the original binary classification formulations and used AUROC as the primary metric, consistent with the paper.

Results — Our findings align with the paper’s primary conclusions. We found the same relative trends: zero-shot detectors outperform pre-trained ones in terms of generalization, expert-edited texts significantly lower detector performance, while LLM-edited texts continue to be more detectable. The repeated results corroborate the original claims, even though the precise AUROC values varied slightly from those stated (usually within a few percentage points).

What was easy — Because the dataset, detectors, and binocular scoring techniques were publically accessible, working with the benchmark was simple. We were able to swiftly replicate a number of key experiments because the evaluation tasks were well explained in the study and most models operated without significant changes.

What was difficult — Because we did not have access to human expert annotators, we were unable to replicate the studies requiring new expert changes. Furthermore, limited access to the particular proprietary models utilized in the article made it impossible to replicate several LLM-based editing workflows. We were only able to analyze a limited number of setups since running several detectors was more computationally demanding than anticipated.

Communication with original authors — Since the dataset and experiment details were adequately documented in the publication and its supporting materials, we did not need to get in touch with the authors. There were no blocking issues that needed more explanation.

1. Introduction

Recently, people have started utilizing many large language models (LLMs) for a wide range of tasks like writing, summarizing etc. While LLMs are beneficial, the wide adoption of it has raised concerns regarding the authenticity of the textual content and potential malicious uses in the domains of news, social media, science, and education. For mitigating these risks, a broad range of machine generated texts (MGT) detection benchmarks have been created which prove to be reliable detectors. However, most of these focus on single-author scenarios consisting of only human and machine generated texts but not the cases like where machine generated texts that have been edited by humans or LLMs.

This is where “Beemo” comes into the picture. It is an evaluation benchmark dataset containing machine generated and LLM/human edited texts and is evaluated on multiple detectors. According to the previous work, only three of them comprise machine-generated & LLM-edited texts and only two of them include machine-generated and human-edited texts. Standard MGT detection benchmarks include various task formulations and labelling schemes such as binary classification, neural authorship attribution, boundary-detection and multi author classification. In addition to this there are several approaches for editing generated content such as operation based, prompt based and expert-based editing.

2 Scope of reproducibility

This paper addresses the problem of MGT texts being detected after being edited by humans/LLMs.

The main contributions of the paper are:

- i) Creation of Beemo, one of the first multi-author benchmarks of LLM-generated and expert-edited responses for fine-grained MGT detection, which counts 19.6k texts in total
- ii) The paper evaluates 33 configurations of zero-shot and pretrained MGT detectors
- iii) The paper releases Beemo and all annotation materials.

In our case, we focus on (ii), since (i) requires expert annotators who we do not have access to and (iii) talks about the release of the material.

The claims mentioned by the paper that we aim to reproduce are listed as follows:

Claim 1: The paper observes that overall binoculars are the best performing zero-shot detector while mage is the strongest pre-trained detector in most scenarios. Entropy performs worse or on par with a random guess classifier.

Claim 2: Expert-based editing evades detection - All detectors often fail to recognize an MGT after it is refined by experts, with the AUROC score decreasing by up to 22% for zero-shot and pretrained detectors.

Claim 3: LLM-edited texts are less likely to be recognized as human-authored - Expert-based texts which are Llama 3.1-70B-instruct-edited are more likely to be classified as human-written compared to the LLM-edited ones such as GPT-4o-edited texts.

Claim 4: Analysis by detector type -

The zero-shot detectors:

1. Are better at distinguishing between human-written and machine-generated texts.
2. Are more generalizable to expert and LLM edited texts which is indicated by lower AUROC scores.
3. Perform better with a larger backbone model in the H vs M/E/L/G scenarios, however, this effect is less pronounced in the E/L/G vs M.

Claim 5: Many zero-shot detectors identify LLM-edited texts as MGTs, with Binoculars, Rank, Log-Rank, and Log Probability demonstrating stronger generalization abilities. However, RADAR, AIGC MPU, and OpenAI RoBERTa-base/large exhibit a random guessing performance across various edit types, which suggests a strong distribution shift.

Claim 6: Analysis by category - Detectors behavior w.r.t different edit types remains the same but the AUROC scores across most task formulations depend on category.

Claim 7: The paper analyzes the effect of the edit percentage in the E vs. M scenario to assess the impact of expert edits. The overall trend is that the detectors' performance remains moderate and stable across the experts' edit ratios, showing general improvement as the ratio increases. Most detectors achieve similar AUROC scores for both moderately (20%–40%) and significantly (60%–80%) edited texts. Lower AUROC scores at the lower edit ranges indicate that moderate editing can confuse the detectors.

These claims are evaluated through complete experiments using 33 detection configurations across seven different binary classification task formulations on the Beemo benchmark dataset.

3 Methodology

We followed the implementation strategy used in the original Beemo paper and relied primarily on the authors released code and model resources. The paper uses the codebases from (Hans, 2023) and (Su, 2024) to run zero-shot detectors, as well as publicly available pretrained detectors such as AIGC-MPU, MAGE, RADAR, and OpenAI's RoBERTa detectors. Since these implementations were already provided, we did not re-implement any detector algorithms from scratch. Instead, we used the same scripts and default backbone models (GPT2-XL, OPT-1.3B, Falcon-7B, and Qwen2-7B) as listed in the original paper to ensure consistency.

We closely mirrored the authors' experimental setup in our preprocessing pipeline. This encompassed:

- Tidying and standardizing text in all formats
- Analyzing and arranging the various authored text versions according to the Beemo dataset format.
- Making format uniform, whitespace, tokenization process, and text size prior to submitting inputs to the detectors.

For assessment, we adhered strictly to the benchmark's methodology. We calculated AUROC directly from the continuous probability outputs of the detectors, like the approach in the paper. AUROC was utilized to assess the effectiveness of each detector in distinguishing between human-written and machine-generated text at all potential thresholds. We also calculated accuracy using a median-centered threshold, adhering to the conventional zero-shot approach employed by the authors.

3.1 Model descriptions

The study evaluates a comprehensive set of zero-shot and pretrained detectors for Machine-generated Text (MGT) detection. In total, we implemented 13 zero-shot configurations (Binoculars, 4 detector types x 3 backbone models) and 5 pretrained detectors which totals to 18 models, achieving the comprehensive coverage of most of the models reported in the original benchmark.

Zero shot detectors:

- Log Probability (GPT-2 XL, OPT-1.3B, Falcon-7B) – Classifies text as machine-generated when its average token log probability is high under a reference LLM.
- Log-Rank (GPT-2 XL, OPT-1.3B, Falcon-7B) – Labels text as machine-generated when the average log-rank of its tokens is high.
- Rank (GPT-2 XL, OPT-1.3B, Falcon-7B)– Identifies text as machine-generated when tokens have higher average rank (lower likelihood) in the model's vocabulary.
- Entropy (GPT-2 XL, OPT-1.3B, Falcon-7B) – Treats text with higher average token-level entropy as more likely to be machine-generated.
- Binoculars – Uses the ratio of perplexities from two LLMs to detect machine-generated text via cross-perplexity comparison.

The parameters of the backbones of the models we implemented are 1.5 billion for GPT-2 XL, 1.3 billion for OPT-1.3B and 7 billion for Falcon-7B.

Pretrained detectors

- RADAR - A 7B-parameter pretrained detector trained via adversarial learning, where a paraphraser tries to evade detection and the detector learns from these hard examples.
- AIGC-MPU - A pretrained detector for edited or partial AI-generated content that uses a multiscale positive-unlabeled loss and text multi-scaling to better capture short and mixed-authorship texts.
- MAGE - A Long former-based detector fine-tuned on 447K human and machine texts to improve generalization across styles and domains.
- OpenAI RoBERTa-base A 125M-parameter RoBERTa model fine-tuned by OpenAI for early GPT-2-based machine-generated text detection.
- OpenAI RoBERTa-large - A stronger 355M-parameter RoBERTa variant fine-tuned by OpenAI for more robust AI-text detection.

3.2 Datasets

The Beemo dataset, which comprises 19,683 texts from 2,817, prompts for five tasks - closed QA, open QA, open-ended creation, rewriting, and summarization - is used in the studies. The dataset offers several authorship types for each question, including versions from GTP-4o and Llama-3.1-70B that are human-written(H), machine-generated (M), expert-edited (E), and LLM-edited. In general, machine-generated texts are longer than their human-written and edited counterparts.

Category	# Examples	# Tokens (P)	# Tokens (H)	# Tokens (M)	# Tokens (E)	# Tokens (L)	# Tokens (G)
Closed QA	1,845	268.0 $\pm$ 230.3	26.7 $\pm$ 17.4	83.3 $\pm$ 73.2	60.9 $\pm$ 66.4	74.2 $\pm$ 35.5	62.4 $\pm$ 37.4
Generation	5,265	40.3 $\pm$ 34.3	225.5 $\pm$ 142.9	305.6 $\pm$ 175.4	278.1 $\pm$ 171.4	115.8 $\pm$ 23.1	115.9 $\pm$ 28.5
Open QA	4,347	15.4 $\pm$ 44.9	89.1 $\pm$ 55.0	186.3 $\pm$ 145.4	128.4 $\pm$ 121.5	100.7 $\pm$ 30.2	91.7 $\pm$ 37.7
Rewrite	4,725	296.7 $\pm$ 249.7	240.6 $\pm$ 201.0	250.3 $\pm$ 160.2	242.7 $\pm$ 170.3	115.7 $\pm$ 22.1	112.2 $\pm$ 28.2
Summarize	3,501	274.9 $\pm$ 162.9	73.7 $\pm$ 45.2	143.4 $\pm$ 110.6	98.2 $\pm$ 59.9	102.5 $\pm$ 25.0	85.9 $\pm$ 30.8
Overall	19,683	159.4 $\pm$ 204.0	153.3 $\pm$ 151.6	216.3 $\pm$ 164.0	184.2 $\pm$ 160.4	101.8 $\pm$ 27.2	93.6 $\pm$ 32.5

Fig 1. Overview of the Beemo Dataset Showing How Each Detection Task Category Varies in Sample Count, Average Text Length, and Included Text Variants

Since Beemo is designed as an evaluation benchmark dataset and does not provide train/dev/test splits. It reports the performance of the detectors by evaluating them directly on the dataset.

Preprocessing includes:

- Collecting prompts and human answers from the No Robots dataset.
- Generating machine responses using 10 instruction-tuned open-source LLMs with default HuggingFace templates.
- Expert editors revise machine outputs for correctness, naturalness, and fluency, with low-quality or non-edited items removed.
- Producing LLM-edited versions using GPT-4o and LLama-3.1-70B with prompts encouraging natural, human-like rewriting.
- Automated checks (similarity, edit distance, stylistic metrics) to validate edit quality.

Download Link: <https://huggingface.co/datasets/toloka/beemo>

3.3 Hyperparameters

We did not perform an extensive hyperparameter search because all evaluated methods are zero-shot detectors that rely on fixed pretrained models. Instead, we used the parameters of the backbones recommended in the original implementations of Entropy, Log-Probability, Log-Rank, Rank, and Binoculars. Each detector was run with three backbone models, namely, GPT-2 XL (1.5B), OPT-1.3B, and Falcon-7B. These model choices were fixed and not tuned; all were loaded in inference-only mode with default tokenizer and generation settings from the Hugging Face library.

3.4 Experimental setup and code

The evaluation technique outlined in the publication is followed in our studies. We use the Beemo dataset as an out-of-domain benchmark and evaluate detector generalization across seven binary classification tasks (H vs M, H vs E, H vs L, H vs G, E vs M, L vs M, G vs M). Each task is constructed by pairing human-written text (H), machine-generated text (M), and edited versions produced by experts (E), Llama-3.1-70B-Instruct (L), or GPT-4o (G), with labels assigned exactly as in the paper formulation.

AUROC is the area under the receiver operating characteristic curve. It was the primary evaluation metric for ranking detectors since it is threshold-independent and commonly used in prior detection work. This makes it especially suitable for our detectors, which output continuous scores rather than fixed labels. A detector with AUROC close to 1.0 reliably ranks machine-generated text higher than human text across the entire distribution.

[Link to our code](#)

3.5 Computational requirements

All experiments were run on Google Colab Pro, which provides access to accelerated hardware with a NVIDIA Tesla T4 GPU (16 GB VRAM), A100 GPU, Intel Xeon 2–4 vCPUs, and 25–30 GB RAM. The software environment included Python 3.10, PyTorch 2.1, Transformers 4.41, and CUDA 12.x. Colab Pro offers 100 compute units, and all experiments were

Type of detector	Experiment ran	GPU RUNTIME (mins)
zero shot	binoculars	110
	Log probability	Gpt2xl- 37, opt1.3b- 33, falcon7b- 39
	Log rank	Gpt2xl- 46, opt1.3b- 32, falcon7b- 50
	rank	Gpt2xl- 46, opt1.3b- 31, falcon7b- 48
	entropy	Gpt2xl- 38, opt1.3b- 25, falcon7b- 40
pretrained	radar	11
	Aigc mpu	10
	mage	37
	openai-roberta-base	6
	openai-roberta-large	11

completed well within this limit. We recorded the GPU runtime for each model in the table.

4 Results

BEEMO contains 33 detector configurations out of which we evaluated 18 (about 55% coverage) through all seven binary classification tasks given.

4.1 Results reproducing original paper.

Result 1 acknowledging claim 1: The claim 1 is strongly supported by the results we achieved for binoculars and Mage showing the higher performance among all the detectors and entropy being the least performing detector.

Detector		H vs. M	E vs. M	H vs. E	L vs. M	H vs. L	G vs. M	H vs. G
Zero-shot MGT Detectors								
Binoculars		83.90	76.79	61.24	59.90	79.90	57.93	78.15
Log Probability	GPT2-XL	69.72	64.73	56.56	63.70	58.88	60.32	61.23
	OPT-1.3B	74.82	66.52	60.86	66.58	62.78	61.12	66.95
	Falcon-7B	86.07	65.48	78.41	65.36	80.78	59.65	82.44
	Qwen2-7B	87.77	68.68	78.02	56.23	84.52	62.45	81.27
Rank	GPT2-XL	60.36	56.70	54.50	59.85	50.65	56.79	53.64
	OPT-1.3B	65.95	57.44	59.94	63.56	53.48	57.10	59.86
	Falcon-7B	74.52	56.56	70.66	62.95	63.64	55.57	70.61
	Qwen2-7B	74.22	57.65	69.11	62.55	72.52	55.10	70.81
Log-Rank	GPT2-XL	70.26	64.95	56.78	65.25	57.42	61.20	60.68
	OPT-1.3B	73.03	66.11	59.03	67.21	59.30	61.83	63.90
	Falcon-7B	84.81	65.07	77.02	66.12	78.26	60.26	80.32
	Qwen2-7B	86.91	68.13	77.04	56.04	83.62	62.09	80.33
Entropy	GPT2-XL	38.93	42.42	46.06	36.74	51.72	41.95	46.64
	OPT-1.3B	37.83	42.74	44.18	38.41	51.33	41.07	45.72
	Falcon-7B	20.92	41.65	34.42	38.02	29.24	40.96	25.66
	Qwen2-7B	10.36	32.02	17.74	44.01	12.82	38.01	15.28
DetectLLM Likelihood Log-Rank Ratio	GPT2-XL	68.73	63.72	56.18	67.45	51.86	62.27	57.07
	OPT-1.3B	64.37	63.09	51.90	67.21	47.17	62.49	52.10
	Falcon-7B	76.73	62.16	68.41	66.77	64.68	61.28	68.31
	Qwen2-7B	79.86	64.12	69.89	54.71	76.25	59.41	72.83
DetectLLM Normalized Perturbed Log-Rank	GPT2-XL	64.06	63.07	62.00	54.08	63.34	51.07	63.93
	OPT-1.3B	65.29	63.86	62.86	56.98	64.04	51.36	65.12
	Falcon-7B	70.14	63.18	66.71	56.65	68.40	51.37	69.84
	Qwen2-7B	74.12	68.30	68.11	54.52	72.49	50.68	73.84
DetectGPT	GPT2-XL	64.91	63.71	62.75	59.15	63.53	58.97	63.63
	OPT-1.3B	67.27	66.87	64.22	64.39	64.68	61.65	65.32
	Falcon-7B	70.69	65.35	67.47	64.15	67.73	62.67	68.18
	Qwen2-7B	72.76	69.37	68.21	61.36	70.14	63.05	69.90
Pretrained MGT Detectors								
RADAR		51.41	50.93	50.80	34.51	62.65	37.96	60.16
MAGE		73.72	60.88	64.09	57.95	67.79	60.59	65.37
AIGC-MPU		70.52	70.35	50.66	65.71	57.17	59.66	62.69
OpenAI RoBERTa-base		66.95	66.83	69.13	66.95	50.00	66.95	50.00
OpenAI RoBERTa-large		62.93	63.96	48.76	64.71	48.57	59.67	53.74

Detector		H vs. M	E vs. M	H vs. E	L vs. M	H vs. L	G vs. M	H vs. G
Zero-shot MGT Detectors								
Binoculars		83.91	76.61	61.64	59.89	79.87	57.91	78.17
Log Probability	GPT2-XL	69.65	64.21	57.09	63.56	58.98	60.17	61.31
	OPT-1.3B	71.75	66.15	57.23	66.40	58.62	60.93	63.27
	Falcon-7B	69.52	65.06	56.06	65.20	57.04	59.48	61.87
Rank	GPT2-XL	60.33	56.36	54.76	59.73	50.77	56.65	53.77
	OPT-1.3B	61.54	57.00	55.19	63.43	47.53	56.91	54.84
	Falcon-7B	58.84	56.11	53.15	62.86	45.46	55.43	53.57
Log-Rank	GPT2-XL	70.17	64.46	57.25	65.11	57.49	61.05	60.74
	OPT-1.3B	71.60	65.77	57.48	67.05	57.35	61.65	62.21
	Falcon-7B	69.55	64.69	56.45	65.97	55.96	60.09	61.19
Entropy	GPT2-XL	39.06	42.56	46.09	36.97	51.65	42.19	46.56
	OPT-1.3B	39.06	42.83	45.79	35.58	53.13	41.27	47.35
	Falcon-7B	39.26	41.91	46.91	36.20	53.42	41.17	47.94
Pretrained MGT Detectors								
RADAR		51.37	52.19	49.66	34.51	62.69	37.95	60.17
MAGE		73.19	62.48	61.56	58.31	66.48	59.29	65.90
AIGC-MPU		70.53	67.26	54.11	65.62	57.30	59.60	62.77
OpenAI RoBERTa-base		67.01	66.64	49.50	64.94	52.76	61.08	57.38
OpenAI RoBERTa-large		62.99	63.87	48.88	64.71	48.65	59.69	53.79

Fig 2. Comparison of original and reproduced detector performance on the Beemo benchmark. The left heatmap shows the AUROC scores reported in the original paper, and the right heatmap shows our reproduced results using the same code and dataset. Both follow the same structure and display consistent performance patterns, where higher (green) values indicate stronger human-machine discrimination.

Result 2 acknowledging claim 2: While comparing the scenarios of H vs M/E using AUROC metrics we observed substantial performance drops across all detectors. From our observation we can say that our results strongly support Claim 2 as we notice a consistent drop up-to 22% across both zero-shot and pretrained detectors, claiming that the expert editing likely evades detection.

Detectors	H vs M	H vs E	Performance drop from our results	Performance drop in paper
Binoculars	0.839	0.616	-22.3%	-22%
Log Probability (GPT-2 XL)	0.697	0.571	-12.6%	-13.1%
Log Rank (OPT-1.3B)	0.716	0.574	-14.1%	-14%
MAGE	0.732	0.616	-11.6%	-9.6%
AIGC-MPU	0.705	0.541	-19.8%	-19.8%
OpenAI RoBERTa-base	0.670	0.495	-17.5%	-17.8%

Table 1. Performance drop comparison from H vs M to H vs E. The table shows AUROC scores for selected detectors and the corresponding performance drops in our reproduced results versus the original paper.

Result 3: LLM-edited texts are less likely to be recognized as human-authored (Claim 3)

Our reproduced results strongly support Claim 3. Across multiple detectors, expert-edited texts (E vs M) consistently show higher AUROC scores than LLM-edited (L vs M) and GPT-edited (G vs M) texts, indicating they are more difficult to distinguish from human writing.

The human-vs-edited comparisons further support this pattern: H vs E scores remain close to random (~0.5–0.6) while H vs L and H vs G are substantially higher, showing that LLM-edited outputs still retain detectable machine-like structure. The combined table below summarizes both sets of comparisons across representative detectors.

Detector	E vs M	L vs M	G vs M	H vs E	H vs L	H vs G
Binoculars	0.766	0.599	0.579	0.616	0.799	0.782
Log Probability (GPT-2 XL)	0.662	0.664	0.609	0.560	0.57	0.618

MAGE	0.625	0.583	0.593	0.616	0.665	0.659
AIGC-MPU	0.673	0.656	0.596	0.541	0.573	0.659
OpenAI RoBERTa-base	0.666	0.649	0.611	0.480	0.596	0.610

Table 2. AUROC scores for distinguishing expert-edited (E), LLM-edited (L), and GPT-edited (G) text from machine-generated (M). The table reports detection performance across several representative zero-shot and pretrained detectors.

Result 4: Analysis by detector type (Claim 4)

We evaluate Zero-shot detectors showing partial support for Claim 4 because we do not implement the large backbone model “qwen2-7b” due to computational limitations.

1. Zero-shot detectors against human-written and machine-generated texts (H vs. M)

Detector Types		AUROC	Performance drop from H vs M $\rightarrow$ H vs E
Zero-shot	Binoculars	-22.3%	0.839
	Log rank (OPT1.3B)	-12.6%	0.716
	Log Probability (OPT1.3B)	-14.1%	0.718
Pretrained	MAGE	-11.6%	0.732
	AIGC MPU	-16.4%	0.705
	OpenAI RoBERTa-base	-17.5%	0.670

Table 3. Comparison of zero-shot and pretrained detectors on Human vs Machine (H vs M) classification. AUROC scores show the relative strength of each detector family, with zero-shot models generally outperforming pretrained detectors in this setting and Performance drop from Human vs Machine (H vs M) to Human vs Expert-edited (H vs E) for zero-shot and pretrained detectors. The table highlights how editing reduces detectability, with varying sensitivity across detector types.

2. Model Generalizability

For this we are comparing performance drops from H vs. M $\rightarrow$ H vs. E.

From these observations we notice that MAGE has the lowest performance drop of -11.6% which doesn’t consistently support claim 4, which says zero-shot detectors show "lower AUROC scores". See table 3.

3. Effects of backbone model size

We evaluated the zero-shot models: Log Probability, Log-Rank and rank with three backbone sizes in the H vs M scenario.

Detector	GPT2-XL	OPT-1.3B	Falcon-7B
Log Probability	0.697	0.718	0.695
Log rank	0.702	0.716	0.696
Rank	0.603	0.615	0.588

Table 4. AUROC scores of zero-shot detectors across different backbone model sizes (GPT2-XL, OPT-1.3B, Falcon-7B) in the H vs M setting. The table shows how scaling the backbone impacts detection performance.

Claim 4 tells that model performance improves with larger backbone models, but from the results we observe that there is no clear monotonic improvement with increasing model size and thereby doesn’t support it. However, we did not implement the Qwen2-7b backbone on these models and hence cannot fully justify this claim.

Result 5: Zero-shot detectors identify LLM-edited texts as MGTs, while some pretrained detectors show distribution shift (claim 5)

Our results strongly support the claim that zero-shot detectors identify LLM-edited texts as MGTs as seen in the table below.

detector	H vs L	H vs G	takeaway
binocular	0.799	0.782	Clearly identifies LLM-edited as MGT
Log probability opt1.3b	0.586	0.633	Moderate identification
Log rank opt 1.3b	0.574	0.622	Moderate identification
rank	0.475	0.548	Weaker but above random

Table 5. Zero-shot detector performance in identifying LLM-edited (H vs L) and GPT-edited (H vs G) texts. The table shows that zero-shot detectors, especially Binoculars, strongly classify LLM-edited text as machine-generated.

Our results strongly support the claim that some pretrained detectors show distribution shift due to random guessing performance. For instance, RADAR shows a severe distribution shift for H vs E (0.497), L vs M (0.345), and G vs M (0.380), whereas for openAI RoBERTa-base, H vs E (0.495) shows random performance. Similar performance is shown for openAI RoBERTa-large, H vs E (0.488) and H vs L (0.486). This confirms that specific LLM outputs struggle to generalize to diverse LLMs and editing scenarios in Beemo, while zero-shot maintains a better performance.

Result 6 (claim 6): The paper reports that detector performance varies with each category (generation, summarization, openQA, closedQA and rewriting). We cannot validate this claim as we have only computed the aggregated scores and not the category-specific ones due to the availability of limited compute units and time constraints.

Result 7 (claim 7): The paper claims that the performance remains moderate and stable across edit ratios. This is given when the variation is less than 10% across bins. Thus, this claim is strongly supported by some detectors such as log probability, rank, log-rank, entropy, and MAGE. However, it also improves as the ratio increases like in the case of Binoculars, AIGC MPU, openAI RoBERTa models. The paper also claims that “Most detectors achieve similar AUROC for both moderately (20%-40%) and significantly (60%-80%) edited texts”. This is supported by our findings that show that almost all detectors have similar AUROC except Binocular. The paper states that “Lower AUROC scores at lower edit ranges indicate moderate editing confuses detectors”. Although this seems true, our findings indicate otherwise. Surprisingly, we observe that our detectors give higher percentages at lower ranges (0%-20%) and perform poorly in the middle ranges (40%-60%).

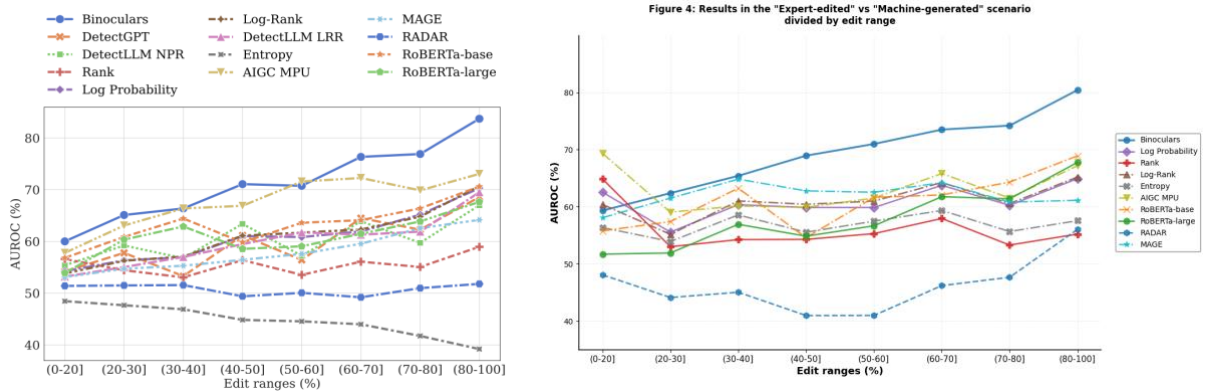


Fig 3. Comparison of detector performance across edit ranges. The left plot shows the original paper's AUROC trends as edit percentage increases, and the right plot shows our reproduced results for the Expert-edited vs Machine-generated scenario, following the same setup and displaying similar overall patterns.

5 Discussion

Our results largely support the main claims of the Beemo paper. Expert-edited texts appeared more human-like than LLM-edited ones and reduced detection accuracy by about 20%, closely matching the original degradation patterns. This was held for both zero-shot and pretrained detectors, confirming claims 2 and 3. Claim 4 received partial support: while zero-shot models like Binocular performed best in the Human vs Machine setting, pretrained models such as MAGE showed smaller AUROC drops and sometimes matched mid-tier zero-shot detectors. We also found no consistent link between model size and performance. Due to compute limits, we could not test claim 6 on category-specific behavior.

Overall, we reproduced 18 of the 33 detectors and all seven binary classification scenarios, achieving AUROC scores closely aligned with the paper. Limited computational resources prevented us from evaluating DetectGPT, DetectLLM, and Qwen2-7B, restricting full analysis of claims 4 and 6. We additionally observed that low edit percentages sometimes produced higher AUROC scores than mid-range edits, contradicting the paper’s finding that moderate edits are most confusing. Finally, while RADAR matched the paper’s AUROC values, its near-random behavior warrants further investigation.

5.1 What was easy.

Reproducing several components of the paper was relatively straightforward due to the availability and clarity of the resources provided. The dataset, model checkpoints, and all essential implementation details—including task formulations, workflow diagrams, and evaluation setups—were publicly accessible. We did not reimplement the detectors ourselves, as the authors’ codebase and referenced repositories allowed us to run each method directly without ambiguity. Additionally, a lot of the jobs eventually boil down to binary classification (human vs. machine or edited vs. original), which makes evaluating and implementing the pipeline simpler. The paper’s workflow diagrams made it easier to properly reconstruct the entire system.

5.2 What was difficult.

Although the reproduction was straightforward to implement, we faced key limitations. Without access to expert annotators, we could not recreate the expert edited MGTs from the original study and instead relied solely on the released dataset. We also did not reproduce the human-in-the-loop editing process or generate new machine-written texts using all 10 LLMs from the paper, which reduced the completeness of our replication.

Computational constraints were another major challenge. Large models used in detectors—such as Qwen2-7B, DetectGPT, and DetectLLM—require significant GPU, CPU, and memory resources, causing slow runtimes and forcing us to downscale batch sizes or process models sequentially. These constraints, combined with small differences in scoring, preprocessing, and backbone versions, made it difficult to match the paper’s exact AUROC values and required extensive debugging to ensure consistency across detectors.

5.3 Communication with original authors

We did not communicate with the original authors during this reproduction study. All necessary resources, including the codebase, datasets, model configurations, and implementation detail were publicly provided in the original paper’s repository and documentation. Since these materials were sufficiently complete for conducting our experiments, no additional clarification or guidance from the authors was required.

6 References

1. Artemova et al. (2025), *Beemo: Benchmark of Expert-edited Machine-generated Outputs*. Proceedings of NAACL 2025.
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3. Su, J., Zhuo, T., Wang, D. and Nakov, P., 2023, December. Detectllm: Leveraging log rank information for zero-shot detection of machine-generated text. In *Findings of the Association for Computational Linguistics: EMNLP 2023* (pp. 12395-12412).